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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 46.7

$$\begin{aligned}
 & \int \frac{x^5 - x^4 + 4x + 4}{x^4 + 4} dx \\
 & \begin{array}{r|l}
 \begin{array}{rrrr}
 x^5 & -x^4 & +4x & +4 \\
 x^5 & +0 & +4x & \\
 \hline
 & -x^4 & +0x & +4 \\
 & -x^4 & +0x & -4 \\
 \hline
 & & & 8
 \end{array}
 & \begin{array}{rr}
 x^4 & +4 \\
 x & 1
 \end{array}
 \end{array} \\
 & = \int (x - 1) dx + \int \frac{8}{x^4 + 4} dx \\
 & = \frac{1}{2}x^2 - x + \int \frac{8}{x^4 + 4} dx \\
 & = \frac{1}{2}x^2 - x + \int \frac{8}{(x^2 - 2x + 2)(x^2 + 2x + 2)} dx \\
 & \Rightarrow \int \frac{8}{(x^2 - 2x + 2)(x^2 + 2x + 2)} dx \\
 & = \int \frac{Ax + B}{x^2 - 2x + 2} dx + \int \frac{Cx + D}{x^2 + 2x + 2} dx \\
 & C(-1 + i) + D = \frac{8}{x^2 - 2x + 2} \Big|_{x=-1+i} \\
 & -C + D + Ci = \frac{8}{(-1 + i)^2 - 2(-1 + i) + 2} \\
 & -C + D + Ci = \frac{8}{1 - 2i - 1 + 2 - 2i + 2} \\
 & -C + D + Ci = \frac{8}{4 - 4i} = \frac{2}{1 - i} = \frac{2(1 + i)}{(1 - i)(1 + i)} = 1 + i \\
 & \Rightarrow C = 1 \\
 & \Rightarrow -C + D = 1 \Rightarrow D = 2 \\
 & A(1 + i) + B = \frac{8}{x^2 + 2x + 2} \Big|_{x=1+i} \\
 & A + B + Ai = \frac{8}{(1 + i)^2 + 2(1 + i) + 2} \\
 & A + B + Ai = \frac{8}{1 + 2i - 1 + 2 + 2i + 2} \\
 & A + B + Ai = \frac{8}{4 + 4i} = \frac{2}{1 + i} = \frac{2(1 - i)}{(1 + i)(1 - i)} = 1 - i \\
 & \Rightarrow A = -1 \\
 & \Rightarrow A + B = 1 \Rightarrow B = 2
 \end{aligned}$$

We lossen eerst de twee integralen apart op.

$$\begin{aligned}
 I_1 &= \int \frac{-x+2}{x^2-2x+2} dx \\
 &= -\frac{1}{2} \int \frac{2x-4}{x^2-2x+2} dx \\
 &= -\frac{1}{2} \int \frac{2x-2-2}{x^2-2x+2} dx \\
 &= -\frac{1}{2} \int \frac{2x-2}{x^2-2x+2} dx + \int \frac{1}{x^2-2x+2} dx \\
 &= -\frac{1}{2} \ln|x^2-2x+2| + \int \frac{1}{(x-1)^2+1} dx \\
 &= -\frac{1}{2} \ln(x^2-2x+2) + \text{Bgtan}(x-1)
 \end{aligned}$$

$$\begin{aligned}
 I_2 &= \int \frac{x+2}{x^2+2x+2} dx \\
 &= \frac{1}{2} \int \frac{2x+4}{x^2+2x+2} dx \\
 &= \frac{1}{2} \int \frac{2x+2+2}{x^2+2x+2} dx \\
 &= \frac{1}{2} \int \frac{2x+2}{x^2+2x+2} dx + \int \frac{1}{x^2+2x+2} dx \\
 &= \frac{1}{2} \ln|x^2+2x+2| + \int \frac{1}{(x+1)^2+1} dx \\
 &= \frac{1}{2} \ln(x^2+2x+2) + \text{Bgtan}(x+1)
 \end{aligned}$$

Totale integraal:

$$\begin{aligned}
 &= \frac{1}{2}x^2 - x + I_1 + I_2 \\
 &= \frac{1}{2}x^2 - x - \frac{1}{2} \ln(x^2-2x+2) + \text{Bgtan}(x-1) + \frac{1}{2} \ln(x^2+2x+2) + \text{Bgtan}(x+1) + C
 \end{aligned}$$

References